THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 12 m/s	v = at.
2	**(a)** 6 kg	m = F/a.
3	**(c)** zero	cos 90 = 0.
4	**(b)** 0.68 m	v/f.
5	**(d)** 5 m/s	100/20.
6	**(c)** 200 N	600/3.
7	**(c)** 9	0.06 x 150.
8	**(c)** greater for the 256 Hz fork	lambda proportional to 1/f.
9	**(d)** 30 m/s	t up = 3 s; u = gt.
10	**(d)** 12 N	Equal to the applied force at the verge of motion.
11	**(b)** 4	Effort arm/load arm.
12	**(b)** quality or timbre	Waveform.
13	**(b)** 100 m	s = (1/2)at squared.
14	**(b)** 6 N	Change in momentum/time.
15	**(d)** 20 m/s	sqrt(2gh).

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	s = 15 x 20 = 300 m (1); Horizontal line; area gives distance (1).	1+1
17	Net force 12 N (1); a = 2.4 m/s squared (1).	1+1
18	W = 200 J (1); KE = 200 J (1).	1+1
19	v = 340 m/s (1); T = 0.05 s; 10 oscillations 0.5 s (1).	1+1
20	t = 2 s (1); v = 20 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	a = 2 m/s squared (1); s = 25 x 5 = 125 m (1); Average velocity 25 m/s (1).	1+1+1
22	a = 5 m/s squared (1); v = 20 m/s (1); s = 40 m (1).	1+1+1
23	Start: KE 225 J (1); At 5 m: PE 100 J, KE 125 J (1); Top: PE 225 J (11.25 m) (1).	1+1+1
24	f = 400 Hz (1); T = 0.0025 s (1); lambda = 0.85 m (1).	1+1+1
	OR: s_A = t squared; s_B = 20t (1); Equal when t = 20 s (1); Distance 400 m (1).	1+1+1
25	Change in momentum 50000 (1); F = 100000 N (1); Weight = 50000 N; force is twice the weight (1).	1+1+1
	OR: At 50 cm: 40 N, MA 2.5 (1); At 1 m: 20 N, MA 5 (1); Longer effort arm gives larger MA (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Maximum velocity 20 m/s (1); Retardation 4 m/s squared (1); Distances 100, 400 and 50 m (1); Total 550 m (1); Trapezium graph (1).	1+1+1+1+1
	OR: a = 0.5 m/s squared (1); Retardation 1.5 m/s squared (1); Time = 20 s (1); Distance = 900/3 = 300 m (1); Second law used (1).	1+1+1+1+1
27	Start: PE 900 J (1); After 1 s: fallen 5 m, KE 100 J, PE 800 J (1); After 2 s: fallen 20 m, KE 400 J, PE 500 J (1); Ground: KE 900 J (v = 30 m/s) (1); Total 900 J (1).	1+1+1+1+1

	OR: Nearer cliff = 170 m (1); Farther cliff = 340 m (1); Distance between cliffs = 510 m (1); At the centre: 255 m each (1); Time = 510/340 = 1.5 s for both echoes (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 20	3.4/0.17.
2	**(b)** 48 km/h	Harmonic mean 2 x 40 x 60/100.
3	**(a)** 60 kg	Mass is constant.
4	**(b)** 120 J	Only vertical displacement counts.
5	**(a)** 0.00588 s	lambda/v.
6	**(a)** 3.14 m/s	2 pi r/T = 12.56/4.
7	**(a)** 8 N	Equals applied force.
8	**(b)** 45 rupees	9 x 5.
9	**(a)** increases	Larger amplitude.
10	**(c)** 40 m	v squared = 2as.
11	**(d)** 100 N	10/0.1.
12	**(c)** 50 N	L/MA.
13	**(a)** half a wavelength	Half cycle.
14	**(a)** uniform acceleration from rest	v proportional to t.
15	**(c)** 500 N	No acceleration.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Change in momentum 20 kg m/s (1); F = 5 N (1).	1+1
17	KE = 450 J (1); F = 450/0.03 = 15000 N (1).	1+1
18	Sound travels faster in helium (1); Frequency of resonance increases, so pitch is higher (1).	1+1
19	No air resistance (1); Acceleration due to gravity is the same for all (1).	1+1
20	Inertia of motion tends to move in a straight line (1); Car turns under him (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	t = d/v (1); 3400/340 (1); 10 s (1).	1+1+1
22	40 + 0 + 40 = 80 m (1); Time 20 s (1); Average speed 4 m/s (1).	1+1+1
23	Change in momentum per second = 2 x 5 (1); = 10 N (1); Newton's second law (1).	1+1+1
24	KE is lost on impact (1); Converts to heat, sound, deformation (1); Energy is conserved overall (1).	1+1+1
	OR: Megaphone: conical tube directs sound (1); Hearing aid amplifies sound (1); Converts to electric signals and back (1).	1+1+1
25	Time = 2.5 s (1); H = 625/20 = 31.25 m (1); v at 1 s = 15 m/s (1).	1+1+1
	OR: Change in momentum 6 kg m/s (1); F = 300 N for 0.02 s (1); F = 30 N for 0.2 s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Speed 20 m/s (1); Reaction distance 10 m (1); Braking distance 40 m (1); Total 50 m (1); Just stops at the obstacle (1).	1+1+1+1+1
	OR: Surface irregularities (1); Depends on nature and normal force (1); f = 60 N (1); Net 20 N (1); a = 1 m/s squared (1).	1+1+1+1+1
27	MA = 3 (1); Efficiency = 75 per cent (1); Output work = 600 J (1); Input = 800 J (effort moves 8 m) (1); Loss = 200 J (1).	1+1+1+1+1

	OR: A is taller (1); B has more waves in the same time (1); A louder (1); B higher pitch (1); Wavelengths: B is half of A (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 36 per cent	(5 - 3.2)/5.
2	**(d)** decibels	dB.
3	**(b)** 2r	Diameter.
4	**(b)** 50 N	Balanced.
5	**(c)** 5 m/s	v = sqrt(2KE/m).
6	**(d)** 343 m/s	f lambda.
7	**(d)** 2.5 m/s squared	72 km/h = 20 m/s; a = 20/8.
8	**(a)** 4 N	a = 2.
9	**(b)** 500 W	15000/30.
10	**(d)** becomes higher	Higher frequency.
11	**(c)** 5 m/s squared	20/4.
12	**(a)** 600 N	m(g + a).
13	**(a)** 3	L/E.
14	**(a)** time period	T.
15	**(a)** pi r	Half the circumference.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	MA = 4 (1); Effort = 100 N (1).	1+1
17	Measure the time interval between flash and bang at a known distance (1); v = distance/time (1).	1+1
18	a = 2 m/s squared (1); s = 9 x 3 = 27 m (1).	1+1
19	Moon: 119 N (1); Earth: 700 N (1).	1+1
20	W = 500 J (1); P = 100 W (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	W = 100000 J (1); P = 2500 W (1); mgh (1).	1+1+1
22	Depth = 1500 m (1); Double depth: 3000 m (1); Time = 4 s (1).	1+1+1
23	a = 3 m/s squared (1); Area of triangle = 0.5 x 6 x 18 (1); = 54 m (1).	1+1+1
24	Pushing adds a downward component on the roller (1); More normal force, more friction (1); Pulling lifts slightly, reduces friction (1).	1+1+1
	OR: Work in = work out (1); E x 5 = 300 x 1 (1); E = 60 N; MA = 5 (1).	1+1+1
25	lambda = 340/50000 = 6.8 mm (1); Time = 3.4/340 (1); = 0.01 s (1).	1+1+1
	OR: Velocity changes uniformly (1); Average v = (u + v)/2 (1); s = average velocity x t (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(a) 3 s (1); (b) 45 m (1); (c) 6 s (1); (d) 30 m/s downward (1); Symmetry (1).	1+1+1+1+1
	OR: p bullet = 8 kg m/s (1); Gun gets equal and opposite momentum (1); v = 8/4 = 2 m/s (1); Recoil is opposite to the bullet (1); Third law (1).	1+1+1+1+1
27	PE of water to KE to electricity (1); Power = 5000 x 10 x 40 = 2 x 10^6 W (1); Useful power = 1.6 x 10^6 W (1); Efficiency 80 per cent (1); Renewable (1).	1+1+1+1+1

	OR: Pinna collects sound (1); Auditory canal and eardrum (1); Ossicles amplify (1); Cochlea and nerve to the brain (1); Precautions: avoid loud noise, no sharp objects (1).	1+1+1+1+1

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